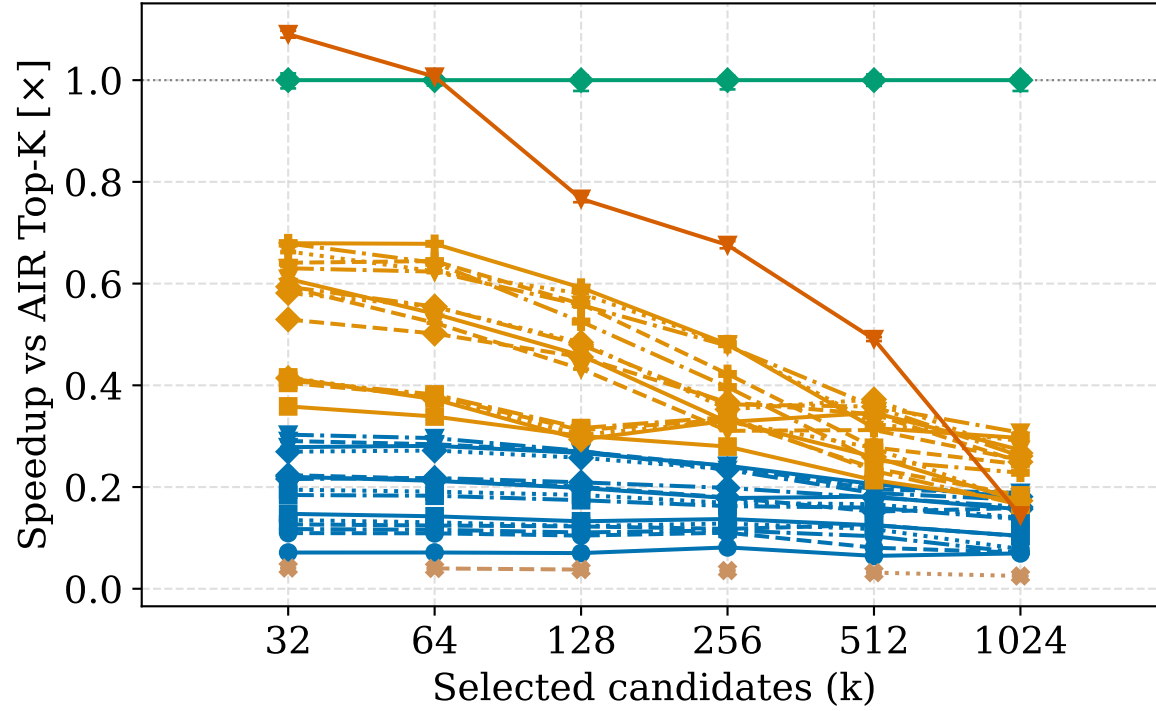
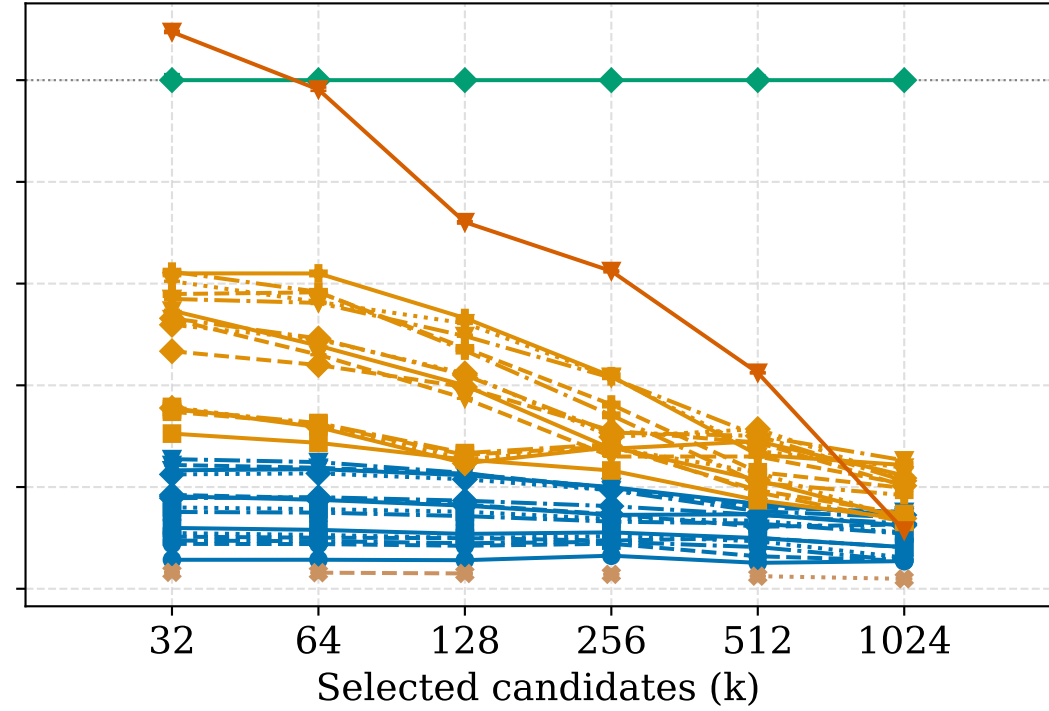


gradient-compression: 589,824 candidates; batch=1; input 773ccccc25; split degree=8
 gradient-compression-small — AIR median / backend median; >1 is faster
 Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



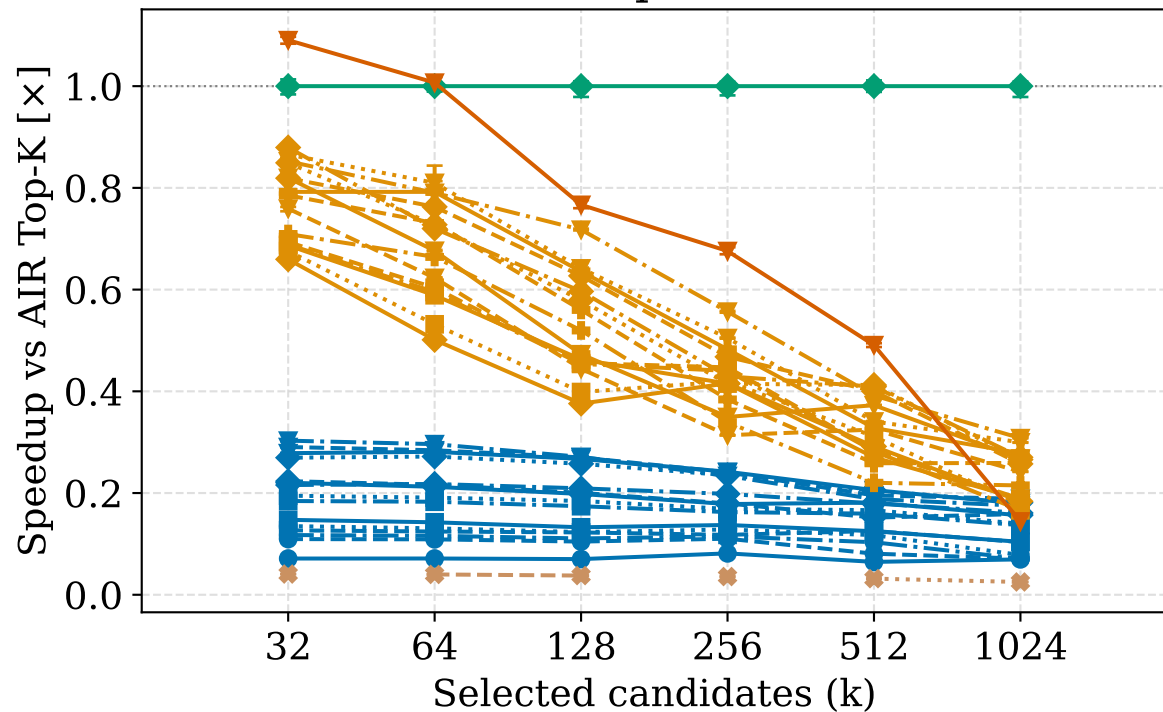
- | | |
|---|--|
| bits (degree=1, block=128, items=4) | bits (split) (degree=8, block=128, items=13) |
| bits (degree=1, block=128, items=7) | bits (split) (degree=8, block=128, items=16) |
| bits (degree=1, block=128, items=8) | bits (split) (degree=8, block=256, items=4) |
| bits (degree=1, block=128, items=13) | bits (split) (degree=8, block=256, items=7) |
| bits (degree=1, block=128, items=16) | bits (split) (degree=8, block=256, items=8) |
| bits (degree=1, block=256, items=4) | bits (split) (degree=8, block=256, items=13) |
| bits (degree=1, block=256, items=7) | bits (split) (degree=8, block=256, items=16) |
| bits (degree=1, block=256, items=8) | bits (split) (degree=8, block=512, items=4) |
| bits (degree=1, block=256, items=13) | bits (split) (degree=8, block=512, items=7) |
| bits (degree=1, block=256, items=16) | bits (split) (degree=8, block=512, items=8) |
| bits (degree=1, block=512, items=4) | bits (split) (degree=8, block=512, items=13) |
| bits (degree=1, block=512, items=7) | bits (split) (degree=8, block=512, items=16) |
| bits (degree=1, block=512, items=8) | AIR Top-K (degree=1, block=512, items=0) |
| bits (degree=1, block=512, items=13) | GridSelect (degree=1, block=0, items=0) |
| bits (degree=1, block=512, items=16) | BlockSelect (degree=1, block=128, items=2) |
| bits (split) (degree=8, block=128, items=4) | BlockSelect (degree=1, block=128, items=3) |
| bits (split) (degree=8, block=128, items=7) | BlockSelect (degree=1, block=128, items=4) |
| bits (split) (degree=8, block=128, items=8) | BlockSelect (degree=1, block=128, items=8) |

gradient-compression: 589,824 candidates; batch=1; input 773ccccc25; split degree=32

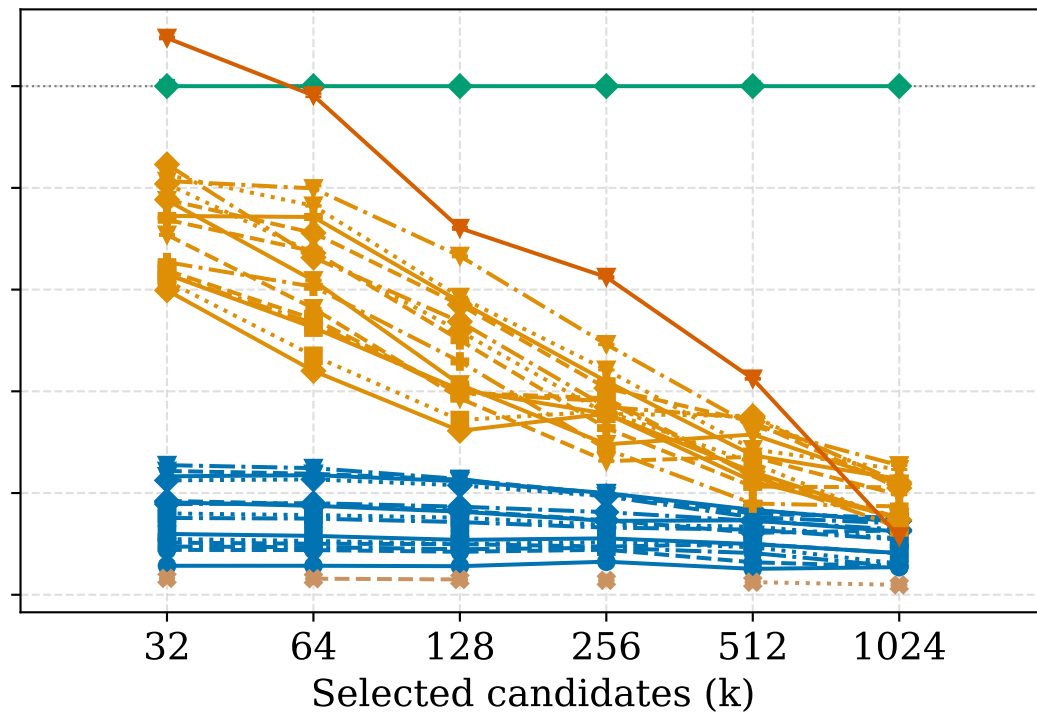
gradient-compression-small — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



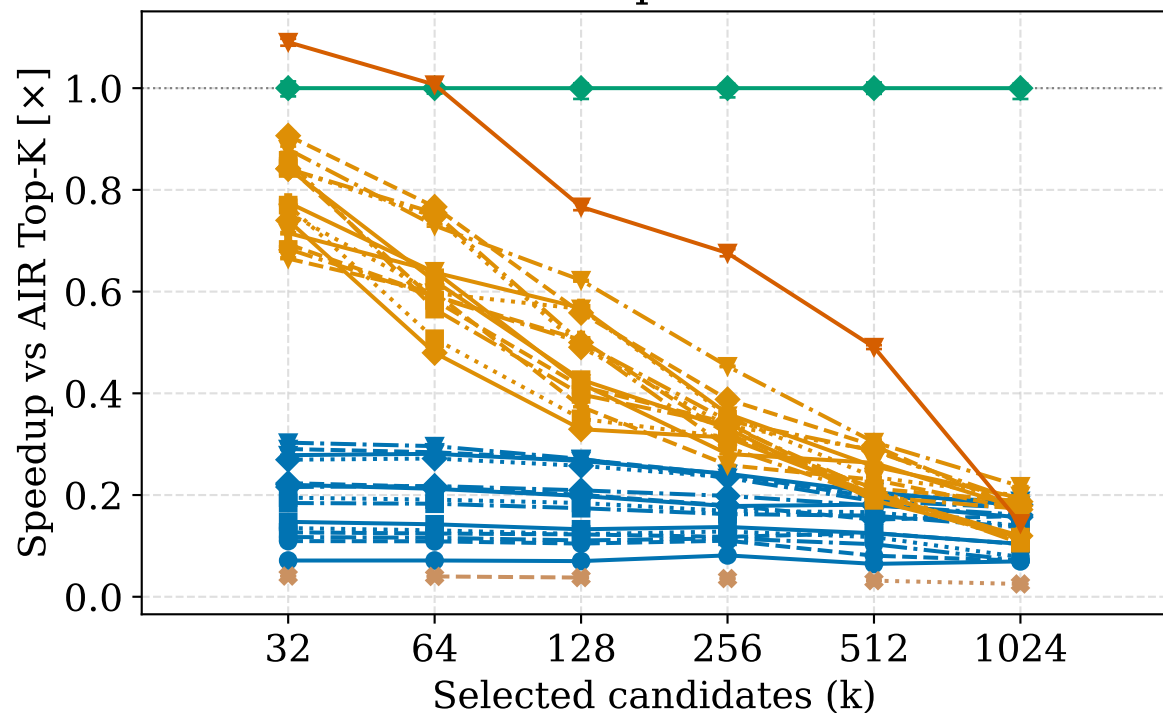
- bits (degree=1, block=128, items=4)
- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=32, block=128, items=4)
- bits (split) (degree=32, block=128, items=7)
- bits (split) (degree=32, block=128, items=8)
- bits (split) (degree=32, block=128, items=13)
- bits (split) (degree=32, block=128, items=16)
- bits (split) (degree=32, block=256, items=4)
- bits (split) (degree=32, block=256, items=7)
- bits (split) (degree=32, block=256, items=8)
- bits (split) (degree=32, block=256, items=13)
- bits (split) (degree=32, block=256, items=16)
- bits (split) (degree=32, block=512, items=4)
- bits (split) (degree=32, block=512, items=7)
- bits (split) (degree=32, block=512, items=8)
- bits (split) (degree=32, block=512, items=13)
- bits (split) (degree=32, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)

gradient-compression: 589,824 candidates; batch=1; input 773cccc25; split degree=128

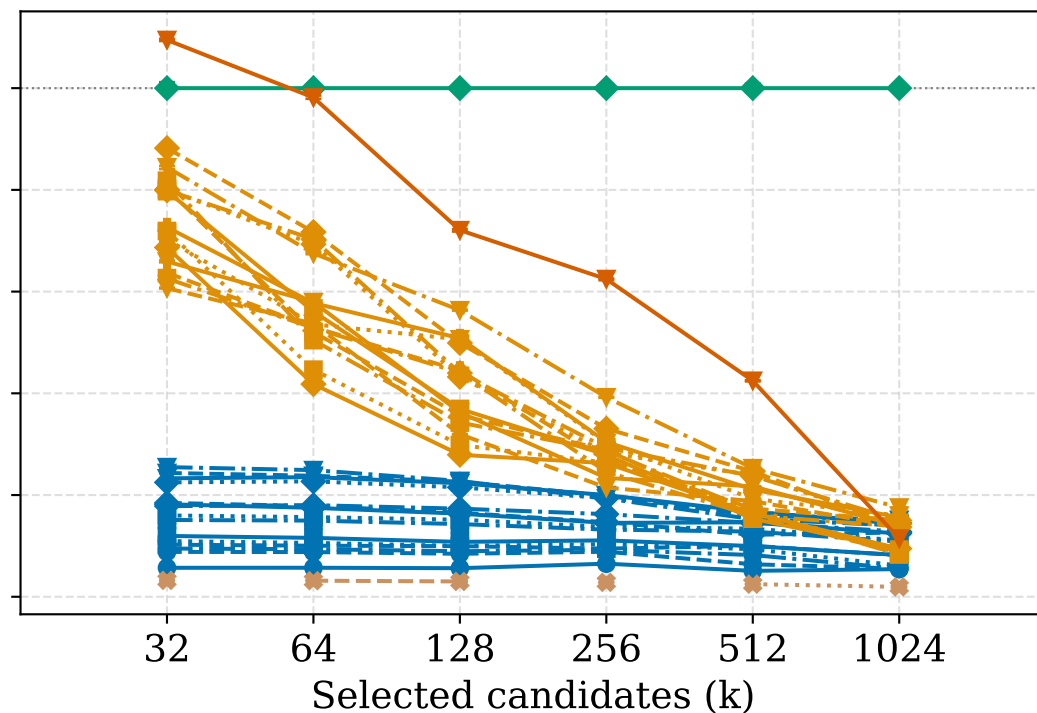
gradient-compression-small — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



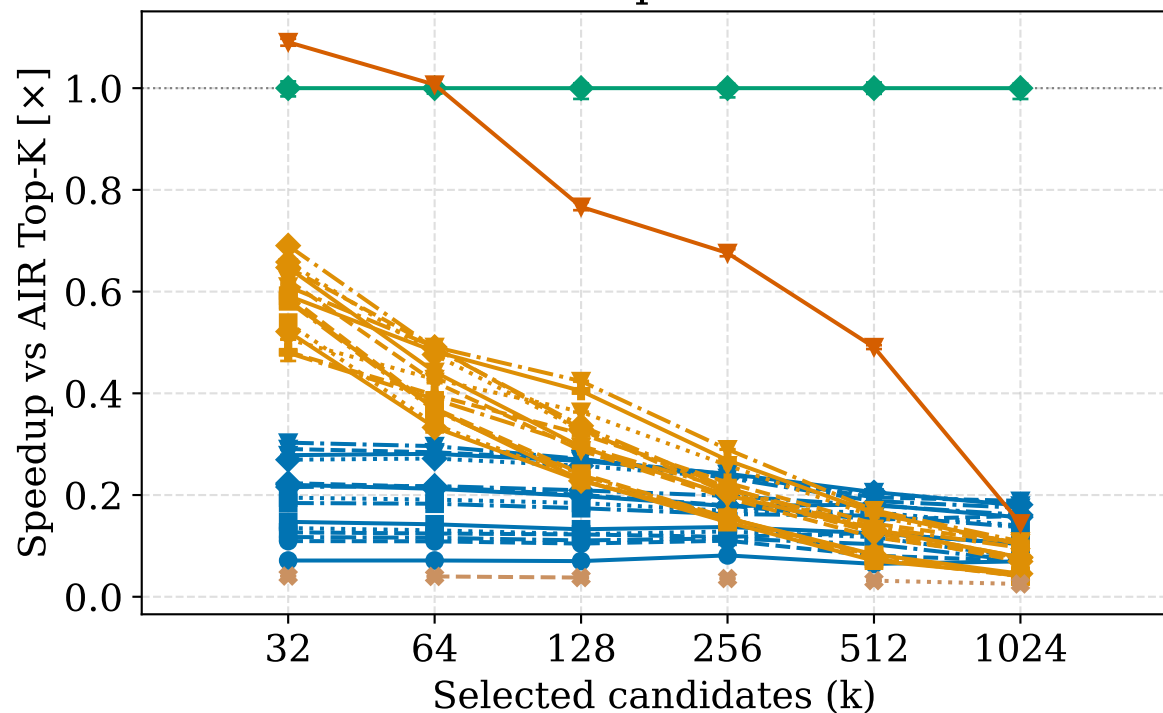
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- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=128, block=128, items=4)
- bits (split) (degree=128, block=128, items=7)
- bits (split) (degree=128, block=128, items=8)
- bits (split) (degree=128, block=128, items=13)
- bits (split) (degree=128, block=128, items=16)
- bits (split) (degree=128, block=256, items=4)
- bits (split) (degree=128, block=256, items=7)
- bits (split) (degree=128, block=256, items=8)
- bits (split) (degree=128, block=256, items=13)
- bits (split) (degree=128, block=256, items=16)
- bits (split) (degree=128, block=512, items=4)
- bits (split) (degree=128, block=512, items=7)
- bits (split) (degree=128, block=512, items=8)
- bits (split) (degree=128, block=512, items=13)
- bits (split) (degree=128, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)

gradient-compression: 589,824 candidates; batch=1; input 773cccc25; split degree=512

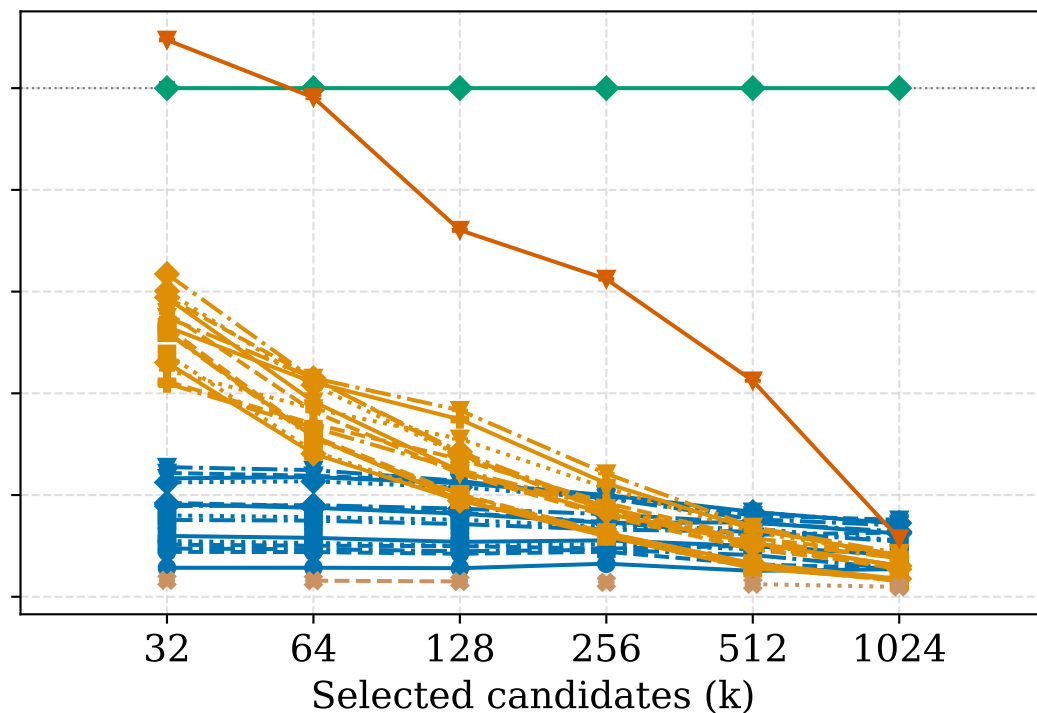
gradient-compression-small — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



- | | |
|---|--|
| bits (degree=1, block=128, items=4) | bits (split) (degree=512, block=128, items=13) |
| bits (degree=1, block=128, items=7) | bits (split) (degree=512, block=128, items=16) |
| bits (degree=1, block=128, items=8) | bits (split) (degree=512, block=256, items=4) |
| bits (degree=1, block=128, items=13) | bits (split) (degree=512, block=256, items=7) |
| bits (degree=1, block=128, items=16) | bits (split) (degree=512, block=256, items=8) |
| bits (degree=1, block=256, items=4) | bits (split) (degree=512, block=256, items=13) |
| bits (degree=1, block=256, items=7) | bits (split) (degree=512, block=256, items=16) |
| bits (degree=1, block=256, items=8) | bits (split) (degree=512, block=512, items=4) |
| bits (degree=1, block=256, items=13) | bits (split) (degree=512, block=512, items=7) |
| bits (degree=1, block=256, items=16) | bits (split) (degree=512, block=512, items=8) |
| bits (degree=1, block=512, items=4) | bits (split) (degree=512, block=512, items=13) |
| bits (degree=1, block=512, items=7) | bits (split) (degree=512, block=512, items=16) |
| bits (degree=1, block=512, items=8) | AIR Top-K (degree=1, block=512, items=0) |
| bits (degree=1, block=512, items=13) | GridSelect (degree=1, block=0, items=0) |
| bits (degree=1, block=512, items=16) | BlockSelect (degree=1, block=128, items=2) |
| bits (split) (degree=512, block=128, items=4) | BlockSelect (degree=1, block=128, items=3) |
| bits (split) (degree=512, block=128, items=7) | BlockSelect (degree=1, block=128, items=4) |
| bits (split) (degree=512, block=128, items=8) | BlockSelect (degree=1, block=128, items=8) |